DATA ITEM DESCRIPTION

Title: HAZARD TRACKING SYSTEM (HTS) DATA

Number: DI-SAFT-82085
AMSC Number: F9742
DTIC Applicable: No
Preparing Activity: 40
Applicable Forms: DD Form 1423

Approved Date: 20161109
Limitation: N/A
GIDEP Applicable: No
Project Number: SAFT-2016-006

Use/Relationship: The Hazard Tracking System (HTS) Data will be used to document and track essential information from contractors for accomplishing the required closed-loop environment, safety, and occupational health (ESOH) hazard tracking processes.

This Data Item Description (DID) contains the format, content, and intended use information for the data deliverable resulting from the work tasks described in the General Requirements, Section 4, of MIL-STD-882, *System Safety*, and is applicable to all developmental and sustaining engineering activities for Department of Defense (DoD) systems, equipment, and facilities. This DID also contains additional data elements required when MIL-STD-882 Detailed Requirements Tasks are applied to the contract.

(Copies of this document are available online at http://quicksearch.dla.mil.)

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The HTS shall be in a format useful and appropriate for the program (e.g., software tools such as spreadsheets or databases). Databases should have relational database capabilities and be presented in a Graphical User Interface.

3. Content. The HTS shall contain the following data elements for each hazard record:

a. Identified hazards: Identify any real or potential conditions that could lead to an unplanned event or series of events (i.e., mishaps) resulting in death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

b. Associated mishaps: Identify the event or series of events resulting in unintentional death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment (includes negative environmental impacts from planned events) related to the hazard.

c. Initial risk assessments: The potential level of risk, comprised of one Severity Category and one Probability Level, based upon initial understanding before any mitigation measures have been taken. Note that this Risk Assessment Code (RAC) never changes-it remains constant throughout the life of the system. There will be multiple event risks across the life cycle of a system.

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d. Target risk assessments: The projected level of risk, comprised of one Severity Category and one Probability Level, projected to be achieved after all mitigations have been implemented.

e. Event(s) risk assessments: The level of risk, comprised of one Severity Category and one Probability Level, associated with a hazard as it applies to a specified hardware/software configuration during an event(s) (e.g., Developmental Testing, Operational Testing, demonstrations, fielding, post-fielding tests).

NOTE: Reporting of the current risk level at a formal technical or program review should be documented as an event risk.

f. Identified risk mitigation measures: Identify the actions considered to eliminate the hazard or reduce the risk.

g. Selected mitigation measures: Identify the actions taken to eliminate the hazard or reduce the risk.

h. Hazard status: Enter the current status of the hazard (e.g., current risk level, formal acceptance of current risk level, planned mitigations or not, etc.).

i. Risk acceptances decisions, in accordance with Department of Defense Instruction (DoDI) 5000.02, *Operation of the Defense Acquisition System*.

NOTE: A single system may require multiple event risk assessments and acceptances throughout its life cycle. Each risk acceptance decision shall be provided with the HTS data.

(Copies of this document are available on line at http://www.dtic.mil/whs/directives/index.html.)

j. Expected risk reduction(s) of alternative(s) estimates: Address expected RAC of each potential risk reduction activity as they reduce the risk from the Initial RAC to the Target RAC.

k. Verification of the implementation of all selected risk mitigation measures: Identify that the mitigations were implemented for selected risk mitigation measures.

l. Validation of all selected risk mitigation measure(s): Identify that the correct mitigation measure(s) was (were) chosen to reduce the risk to the Target RAC.

m. All software contributions to system risk. For software involved with an identified hazard, this contribution is comprised of the Software Control Category (SCC), Software Criticality Index (SwCI), the Level of Rigor (LOR), and the assigned risk level.

4. Additional Requirements/Features. The HTS shall include the following data elements for each hazard if the listed additional MIL-STD-882 Detailed Requirements Tasks are specified in the contract:

4.1 Task 101 HAZARD IDENTIFICATION AND MITIGATION EFFORT USING THE SYSTEM SAFETY METHODOLOGY:

a. Subcontractor, vendor and supplier hazard tracking data: Incorporate the associated data elements from Task 101 and the general requirements data elements in section 3 of this DID for subcontractor(s), vendor(s), and suppliers' hazards.

b. Hazard: List any real or potential conditions that could lead to an unplanned event or series of events (i.e., mishaps) resulting in death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

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c. System: A description of the system where the hazard exists (e.g., Joint Light Tactical Vehicle (JLTV), F-22, Joint Strike Fighter (JSF/F-35), Littoral Combat Ship (LCS)).

d. Subsystem (if applicable): A description of the subsystem(s) where the hazard exists.

e. Applicability: Version specific hardware designs or configurations, software releases.

f. Requirements references: Identify Safety Significant requirements leveraged on the hazard by reviewing DoD and industry standards and specifications; historical documentation on similar and legacy systems; DoD requirements (to include risk mitigation technology requirements); system performance specifications; other system design requirements and documents; applicable Federal, State, and local regulations; and applicable Executive Orders (EOs) and international agreements.

g. System mode(s): Addresses which applicable mode(s) the hazard exists in, to include operational mode(s), maintenance, test, storage, transport, and demilitarization.

NOTE: The system response may differ depending upon the mode of operation.

h. Causal factor: Addresses the mechanism(s) that trigger the hazard (e.g., hardware, software, human, operational environment, natural environment).

i. Effects: The expected outcome if the mishap occurs (e.g., personnel death/injury, environmental impact, monetary loss).

j. Mishap: Identify the event or series of events resulting in unintentional death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment (includes negative environmental impacts from planned events) related to the hazard.

k. Initial risk assessments: The potential level of risk, comprised of one Severity Category and one Probability Level, based upon initial understanding before any mitigation measures have been taken. Note that this RAC never changes - it remains constant throughout the life of the system. There will be multiple event risks across the life cycle of a system.

l. Target risk assessments: The projected level of risk, comprised of one Severity Category and one Probability Level, projected to be achieved after all mitigations have been implemented.

m. Event(s) risk assessments: The level of risk, comprised of one Severity Category and one Probability Level, associated with a hazard as it applies to a specified hardware/software configuration during an event(s) (e.g., Developmental Testing, Operational Testing, demonstrations, fielding, post-fielding tests).

NOTE: Reporting of the current risk level at a formal technical or program review should be documented as an event risk.

n. Mitigation measures: Identify the actions considered to eliminate the hazard or reduce the risk.

o. Mitigation measure(s) status: Describe the status and progress of the effort to implement, verify and validate the selected mitigation measure(s). Include all relevant information on how and when the mitigation will be implemented and whether full incorporation/validation of mitigation measures is tied to a specific milestone, e.g., a specific test, Critical Design Review (CDR), etc.

p. Hazard status: Enter the current status of the hazard (e.g., current risk level, formal acceptance of current risk level, planned mitigations or not, etc.).

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q. Verification and validation method: Address how program plans to verify and validate the effectiveness of planned mitigations to determine if the target risk level had been achieved.

r. Action person(s) and organizational element: Individual or organization within a program (government and contractor) that is responsible for ESOH risk management.

s. Record of risk acceptance(s): Identify the Risk acceptance authority as specified in DoDI 5000.02 (and User Representative Concurrence authority, as applicable) by title and organization, date of acceptance, and location of the accepted risk document(s).

t. Hazard management log (record of hazard entry and changes made to any part of the hazard during the system's life-cycle).

4.2 Task 106 HAZARD TRACKING SYSTEM:

a. Hazard: List any real or potential conditions that could lead to an unplanned event or series of events (i.e., mishaps) resulting in death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

b. System: A description of the system where the hazard exists (e.g., JLTV, F-22, JSF, LCS).

c. Subsystem (if applicable): A description of the subsystem(s) where the hazard exists.

d. Applicability: Version specific hardware designs or configurations, software releases.

e. Requirements references: Identify Safety Significant requirements leveraged on the hazard by reviewing DoD and industry standards and specifications; historical documentation on similar and legacy systems; DoD requirements (to include risk mitigation technology requirements); system performance specifications; other system design requirements and documents; applicable Federal, State, and local regulations; and applicable EOs and international agreements.

f. System mode(s): Addresses which applicable mode(s) the hazard exists in, to include operational mode(s), maintenance, test, storage, transport, and demilitarization.

NOTE: The system response may differ depending upon the mode of operation.

g. Causal factor: Addresses the mechanism(s) that trigger the hazard (e.g., hardware, software, human, operational environment, natural environment).

h. Effects: The expected outcome if the mishap occurs (e.g., personnel death/injury, environmental impact, monetary loss).

i. Mishap: Identify the event or series of events resulting in unintentional death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment (includes negative environmental impacts from planned events) related to the hazard.

j. Initial risk assessments: The potential level of risk, comprised of one Severity Category and one Probability Level, based upon initial understanding before any mitigation measures have been taken. Note that this RAC never changes - it remains constant throughout the life of the system. There will be multiple event risks across the life cycle of a system.

k. Target risk assessments: The projected level of risk, comprised of one Severity Category and one Probability Level, projected to be achieved after all mitigations have been implemented.

l. Event(s) risk assessments: The level of risk, comprised of one Severity Category and one Probability Level, associated with a hazard as it applies to a specified hardware/software

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configuration during an event(s) (e.g., Developmental Testing, Operational Testing, demonstrations, fielding, post-fielding tests).

NOTE: Reporting of the current risk level at a formal technical or program review should be documented as an event risk.

m. Mitigation measures: Identify the actions considered to eliminate the hazard or reduce the risk.

n. Mitigation measure(s) status: Describe the status and progress of the effort to implement, verify and validate the selected mitigation measure(s). Include all relevant information on how and when the mitigation will be implemented and whether full incorporation/validation of mitigation measures is tied to a specific milestone, e.g., a specific test, Critical Design Review (CDR), etc.

o. Hazard status: Enter the current status of the hazard (e.g., current risk level, formal acceptance of current risk level, planned mitigations or not, etc.).

p. Verification and validation method: Address how program plans to verify and validate the effectiveness of planned mitigations to determine if the target risk level had been achieved.

q. Action person(s) and organizational element: Individual or organization within a program (government and contractor) that is responsible for ESOH risk management.

r. Record of risk acceptance(s): Identify the Risk acceptance authority as specified in DoDI 5000.02 (and User Representative Concurrence authority, as applicable) by title and organization, date of acceptance, and location of the accepted risk document(s).

s. Hazard management log: Record of hazard entry and changes made to any part of the hazard during the system's life-cycle).

t. Hazardous Material (HAZMAT) data elements as specified by the Government.

4.3 Task 108 HAZARDOUS MATERIALS MANAGEMENT PLAN:

a. HAZMAT item or substance name (and CAS Number if available): Identify all Prohibited, Restricted and Tracked HAZMATs.

b. HAZMAT Category: For each HAZMAT hazard record, identify whether the HAZMAT is categorized as Prohibited, Restricted, or Tracked.

c. Special Material Content Code (SMCC): Identify the SMCC for the HAZMAT as designated by DoD 4100.39-M, Vol 10, *FLIS Procedures Manual Multiple Application References/Instructions/Tables and Grids*.

(Copies of this document are available online at http://www.dtic.mil/whs/directives/index.html.)

d. Location(s) of the HAZMAT within the system: Identify each location on the system where the HAZMAT appears; with traceability, as applicable, to version specific hardware designs.

e. Quantity of HAZMAT within the system: Identify the quantity of the HAZMAT each identified location on the system where the HAZMAT appears, with traceability, as applicable, to version specific hardware designs.

f. How HAZMAT is embedded in the system: Application, process or activity whereby quantities of HAZMAT are embedded in the system or used during operations, and support of the system.

g. How HAZMAT is used during operations and maintenance: Reasonably anticipated HAZMAT (whether categorized or not categorized) generated during the system's life-cycle

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(e.g., installation, Government test and evaluation, normal use, and maintenance or repair of the system).

h. Name(s) of HAZMAT(s) generated over system's life cycle: Reasonably anticipated HAZMAT (whether categorized or not) generated during the system's life-cycle (e.g., installation, Government test and evaluation, normal use, and maintenance or repair of the system).

i. HAZMAT mitigation measure(s) based on the specific HAZMAT: Special HAZMAT control, training, handling measures, and PPE needed, including provision of required MSDSs.

4.4 Task 201 PRELIMINARY HAZARD LIST:

a. Identified hazards: Identify any real or potential conditions that could lead to an unplanned event or series of events (i.e., mishaps) resulting in death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.

b. Hazard description: A brief description of the hazard shall include the Source (hazardous element), Mechanism (initiating mechanism) and Outcome (threat/target of the damage, injury, or loss).

c. Causal factor: Addresses the mechanism(s) that trigger the hazard (e.g., hardware, software, human, operational environment, natural environment).

4.5 Task 202 PRELIMINARY HAZARD ANALYSIS:

a. Preliminary hazard analysis results.

4.6 Task 204 SUBSYSTEM HAZARD ANALYSIS:

a. Subsystem hazard analysis results.

4.7 Task 205 SYSTEM HAZARD ANALYSIS:

a. System hazard analysis results.

4.8 Task 206 OPERATION AND SUPPORT HAZARD ANALYSIS:

a. Operation and support hazard analysis results.

4.9 Task 207 HEALTH HAZARD ANALYSIS:

a. Health hazard analysis results.

4.10 Task 208 FUNCTIONAL HAZARD ANALYSIS:

a. Functional hazard analysis results.

4.11 Task 209 SYSTEM-OF-SYSTEMS HAZARD ANALYSIS:

a. System-of-systems hazard analysis results.

4.12 Task 210 ENVIRONMENTAL HAZARD ANALYSIS:

a. Environmental hazard analysis results.

b. If hazards are associated with HAZMAT, the following minimum data elements shall be included in the HTS:

(1) HAZMAT item or substance name.

(2) HAZMAT Category (prohibited, restricted, or tracked).

(3) Special Material Content Code (SMCC) as designated in DoD 4100.39-M, Volume 10.

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(4) Location of HAZMAT within the system.

(5) Quantity of HAZMAT within the system with traceability to version specific hardware designs.

(6) Application, process, or activity whereby quantities of HAZMAT are embedded in the system, or used during operations, and support of the system.

(7) Reasonably anticipated HAZMAT (whether categorized or not categorized) generated during the system's life-cycle (e.g., installation, Government test and evaluation, normal use, and maintenance or repair of the system).

(8) Reasonably anticipated HAZMAT (whether categorized or not categorized) generated during mishaps.

(9) Special HAZMAT control, training, handling measures, and PPE needed, including provision of required (MSDSs.

c. If hazards are associated with pollutant (including noise) generation, the following additional data elements shall be included in the HTS:

(1) Identification of the specific pollutants associated with system operations and maintenance activities.

(2) Sources of emission for each pollutant.

(3) Quantity and magnitude or rate of pollution generated during normal operation and maintenance, as specified by the program office.

(4) Special emission control, training, handling measures, and personal protective equipment needed.

(5) Quantity/emission rate of pollutant(s) generated during normal operation and maintenance as specified by the program office: Quantity and magnitude or rate of pollution generated during normal operation and maintenance as specified by the program office.

(6) Pollutant mitigation measure(s) based on the specific Pollutant: Special emission control, training, handling measures, and personal protective equipment needed.

5. Other HTS requirements to be specified in solicitations and contracts:

a. Hazard ID: A unique alpha-numeric hazard identification for each record entry.

b. Originator of hazard: Name, email, and contact number of individual making a change to the data in the HTS.

c. Critical Safety Items (CSIs): If applicable and the hazard affects CSIs, list the affected part numbers.

d. User accounts.

e. Password protection.

f. The ability to embed documents/files.

g. The ability to export reports (e.g., hazard worksheet formats, hazard action reports for risk acceptance) with appropriate Distribution Statements.

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h. Hazard-related system certification requirements traceability (e.g., airworthiness, seaworthiness, nuclear surety, software assurance, etc.).

6. Specific delivery instructions will be specified in Block 14 of the *Contract Data Requirements List (CDRL)*, DD Form 1423.

(Copies of this form are available online at http://www.dtic.mil/whs/directives/forms/index.htm.)

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